

IA na Pesquisa Exemplos de Aplicação



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Conteúdo



Aplicações em Processamento de Linguagem Natural

Aplicações em Processamento de Imagens

Aplicações Híbridas



Aplicações em Processamento de Linguagem Natural

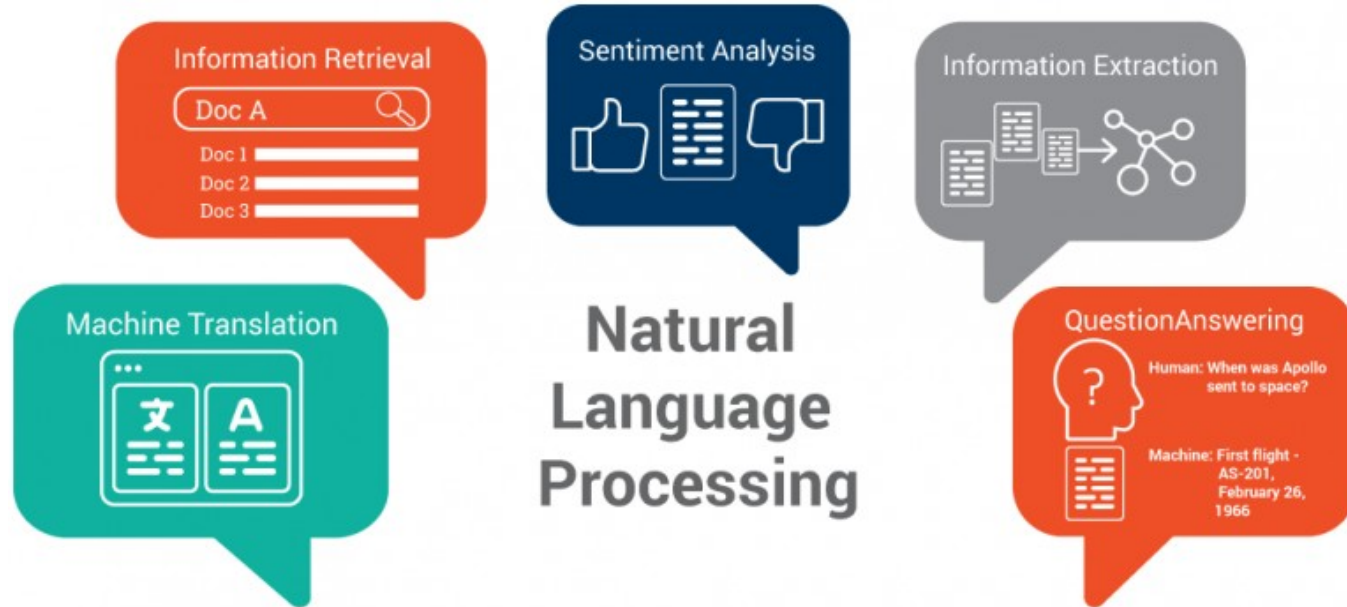
Introdução Proc. de Linguagem Natural



Processamento de Linguagem Natural (PLN) é um campo da Linguística e da Aprendizagem de Máquina focado no entendimento da linguagem humana

Tem como objetivo entender o contexto das palavras

Introdução Proc. de Linguagem Natural



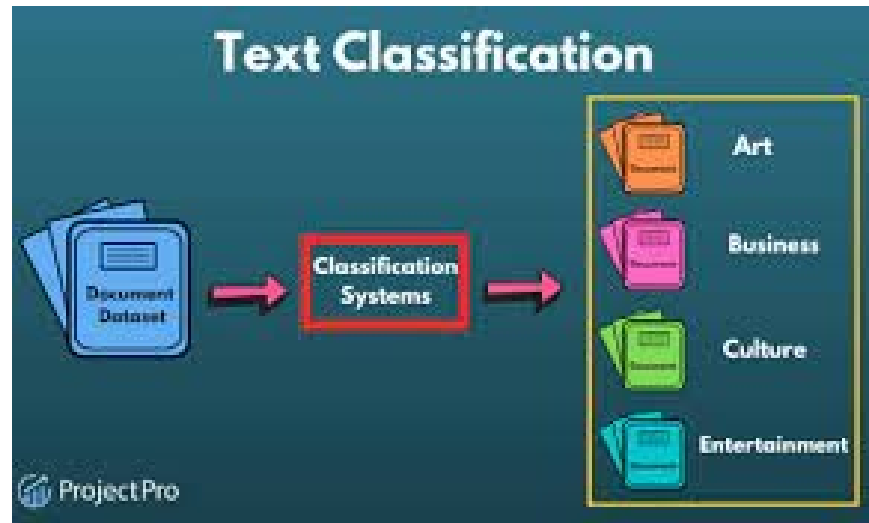
Referência.: <https://www.ontotext.com/blog/top-5-semantic-technology-trends-2017/>
<https://neptune.ai/blog/recurrent-neural-network-guide>

Classificação de Texto

Analisar sentimentos

Detectar spam

Identificar se duas
sentenças estão
relacionadas



Referência.:

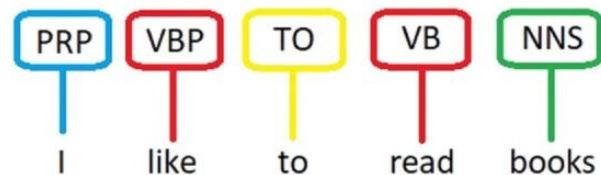
<https://www.projectpro.io/article/machine-learning-nlp-text-classification-algorithms-and-models/523>

Classificação de Palavras

Identificar a classe gramatical
(substantivo, verbo, adjetivo)

Reconhecer entidades (local,
pessoa, organização)

POS Tagging

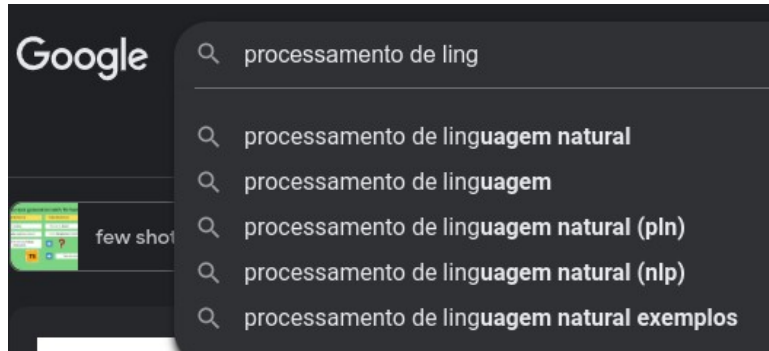


Referência.: <https://byteiota.com/pos-tagging/>

Geração de Conteúdo Textual

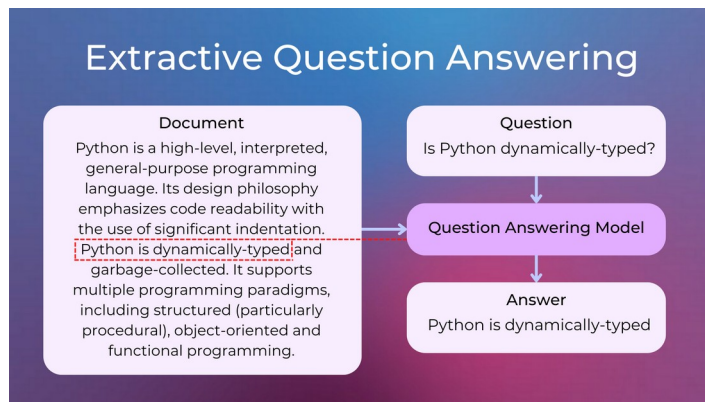
Completar um *prompt*: próxima(s) palavra(s)

Preencher um espaço em branco em um texto com palavras mascaradas



Extração/Geração de Respostas de um Texto

Extrair ou gerar a resposta para uma pergunta dado um contexto



Chatbots

Assistentes Virtuais

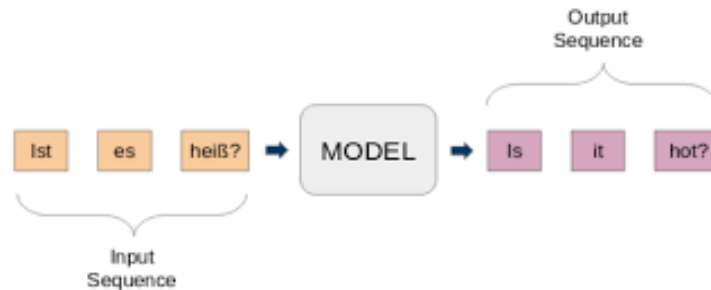
Referência.: <https://www.nlplanet.org/course-practical-nlp/02-practical-nlp-first-tasks/17-question-answering.html>

Geração de Sentenças Novas

Geração de nova sentença a partir de uma entrada

Traduzir um texto para um novo idioma

Sumarizar um texto



Referência.: <https://www.analyticsvidhya.com/blog/2019/01/neural-machine-translation-keras/>

Tarefas mais Complexas

PLN não está limitado à linguagem escrita

Transcrição de áudio

Descrição de uma imagem

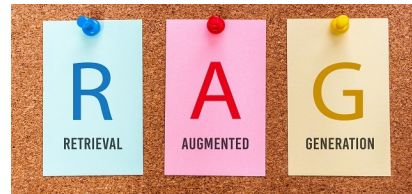
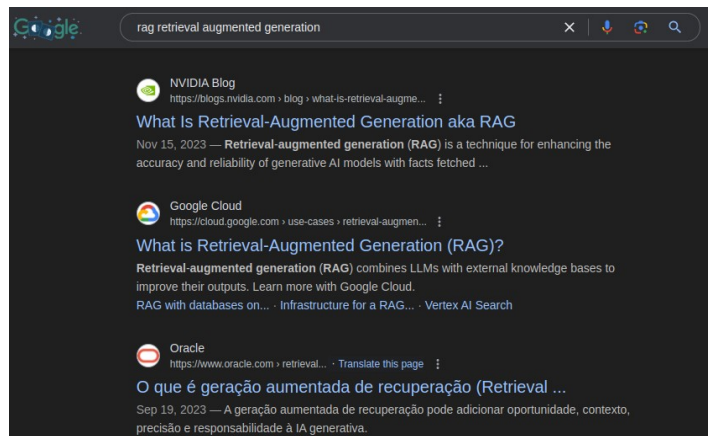


Referência.: <https://www.ics-dubai.com/blogs/what-is-audio-transcription>

Recuperação de Informação

Máquina de Busca

Ranqueia documentos em ordem de relevância para uma consulta

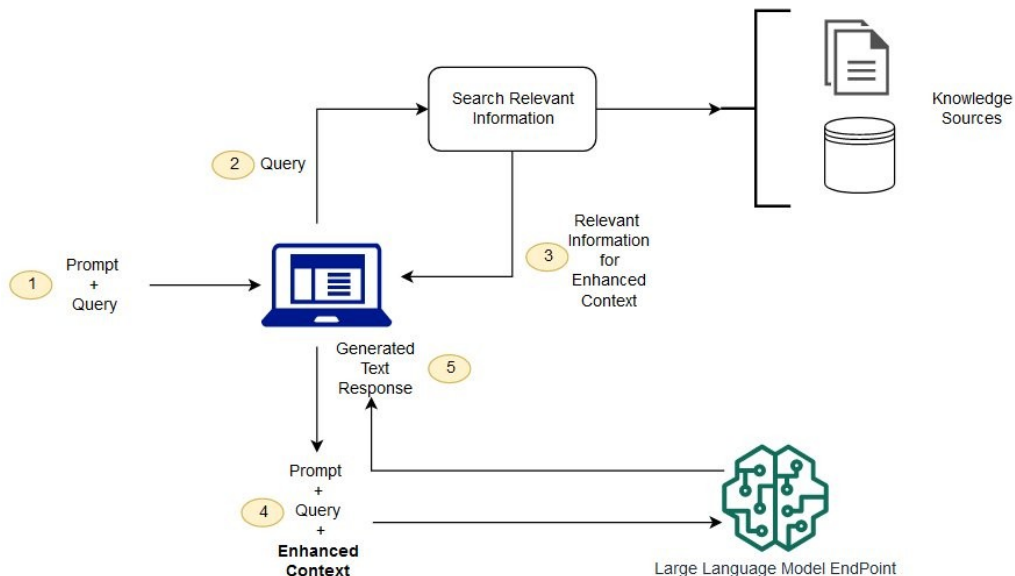


Geração Aumentada (RAG)

Gera respostas para uma consulta usando LLMs

Referência.: <https://opendatascience.com/retrieval-augmented-generation-rag-a-synergistic-approach-to-nlu-and-nlg/>

RAG - *Retrieval Augmented Generation*



Estende os recursos dos LLMs para domínios específicos ou para a base de conhecimento interna de uma organização sem a necessidade de treinar novamente o modelo

Referência.: <https://aws.amazon.com/pt/what-is/retrieval-augmented-generation/>

Análise de Sentimentos



126 31



Avaliou em 14 de Janeiro de 2018 via dispositivo móvel

Tudo de bom

Padaria oferece um excelente café da manhã, mas nada barra o lanche da tarde. Muitas opções , muitas diversidades e pra que aprecia café tem uma boa variedade também! Atendimento rápido e dinâmico.



Obrigado,



5



Avaliou em 4 de Janeiro de 2018 via dispositivo móvel

Bom

Eu encomendei um frango veio meio cru mas tava gostoso, tem varias opcpes de doce tb, gosto muito, menos do frango kkk



Obrigado,

Análise de Sentimentos

(1) Eu comprei um iPhone dois dias atrás. (2) Ele era muito chique. (3) A tela de toque (*touch screen*) era realmente muito legal. (4) A qualidade de voz era muita clara. (5) Entretanto, minha mãe foi muito má comigo porque eu não falei pra ela antes de comprá-lo. (6) Ela também achou que o telefone era muito caro, e queria devolvê-lo para a loja...

Entidades: iPhone, Mãe

Aspectos: tela de toque, qualidade de voz, preço

Sentimentos: positivos, exceto para o aspecto preço e a entidade Mãe

Análise de Sentimentos

Artificial Intelligence Review (2021) 54:1087–1115
<https://doi.org/10.1007/s10462-020-09870-1>



A survey of sentiment analysis in the Portuguese language

Denilson Alves Pereira¹

Published online: 6 July 2020
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Abstract

Sentiment analysis is an area of study that aims to develop computational methods and tools to extract and classify the opinions and emotions expressed by people on social networks, blogs, forums, online shoppings, and others. A lot of research has been developed addressing opinions expressed in the English language. However, studies involving the Portuguese language still need to be advanced to make better use of the specificities of the language. This paper aims to survey the efforts made specifically to address sentiment analysis in the Portuguese language. It categorizes and describes state of the art works involving approaches to each of the tasks of sentiment analysis, as well as supporting language resources such as natural language processing tools, lexicons, corpora, ontologies, and datasets.

Keywords Sentiment analysis · Opinion mining · Natural language processing · Portuguese language

Neural Computing and Applications
<https://doi.org/10.1007/s00521-024-10265-4>

ORIGINAL ARTICLE



Approaches based on language models for aspect extraction for sentiment analysis in the Portuguese language

José Carlos Ferreira Neto¹ · Denilson Alves Pereira² · Bruno Henrique Groenner Barbosa¹ · Danton Diego Ferreira¹

Received: 30 April 2024 / Accepted: 23 July 2024
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Abstract

This work addresses the gap in aspect extraction techniques for Portuguese by adapting methods originally designed for English. It focuses on TV devices and literary reviews in the TV and ReLi datasets. For this, models based on the BERT architecture were employed, including pre-trained general domain (BERTimbau) and specific domain models (BERTtv and BERTreli). Also, this paper contributes with a novel double embedding technique that merges these models. We further explored the potential of large language models (LLMs) with a Portuguese-trained LLaMa variant, Cabrita. Efficient fine-tuning techniques such as LoRA (low-rank adaptation) for BERTimbau and QLoRA (quantized low-rank adaptation) for Cabrita were applied to optimize resource demands. The BERTimbau model, adjusted with LoRA, achieved the highest F1 scores (0.846 for TV and 0.615 for ReLi), while Cabrita showed lower performance (0.68 for TV and 0.46 for ReLi). This study underscores the potential of adapting and optimizing existing techniques for aspect extraction in Portuguese, marking a significant advancement in the field.

Keywords Natural language processing · Aspect extraction · BERT · Language models

Reconhecimento de Entidades Nomeadas

The diagram illustrates Named Entity Recognition (NER) with a legend and a sample text. The legend at the top shows five categories: ORGANISATION (orange), LOCATION (yellow), DATE (green), PERSON (cyan), and WEAPON (blue). Below the legend, a sample text is shown with entities highlighted and labeled with their respective category and type (e.g., ORG, LOC, DATE, PER, WEAPON).

ORGANISATION LOCATION DATE PERSON WEAPON

The **ISIS**_{ORG} has claimed responsibility for a suicide bomb blast in the **Tunisian**_{LOC} capital **earlier this week**_{DATE}, the **militant group**_{ORG}'s **Amaq news agency**_{ORG} said on **Thursday**_{DATE}. A **militant**_{PER} wearing an **explosives belt**_{WEAPON} blew himself up in **Tunis**_{LOC}

Referência.: <https://www.shaip.com/blog/named-entity-recognition-and-its-types/>

Reconhecimento de Entidades Nomeadas

Language Resources and Evaluation
<https://doi.org/10.1007/s10579-023-09665-0>

ORIGINAL PAPER



CachacaNER: a dataset for named entity recognition in texts about the cachaça beverage

Priscilla Silva¹ · Arthur Franco¹ · Thiago Santos¹ · Mozar Brito² · Denilson Pereira¹

Accepted: 3 May 2023
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Abstract

Named Entity Recognition (NER) is the task of identifying and classifying tokens in texts corresponding to a set of pre-defined categories, such as names of people, organizations and locations. Datasets labeled for this task are essential for training supervised machine learning models. Although there are many datasets labeled with texts for English, in the Portuguese language they are scarcer. This work contributes to the creation and evaluation of a manually labeled dataset for the NER task, with texts in Brazilian Portuguese, in the specific domain of the beverage called Cachaça. This is a popular drink in Brazil, and of great economic importance. This is the first NER dataset in the beverage domain, and can be useful for other types of beverages with similar entity categories, such as wine and beer. We describe the process of data collection, creation of the dataset and its experimental evaluation. As a result, we created a dataset containing over 180,000 tokens labeled in 17 entity categories. The labeling obtained an agreement coefficient of 0.857 among the labelers, according to the Fleiss' Kappa metric, which is considered almost perfect. In our experimental evaluation, we obtained a micro-F1 value equal to 0.933 in the test set. The size of the dataset, as well as the result of its experimental evaluation, are comparable to other datasets in the Portuguese language, even though ours has a greater number of entity categories.

Keywords NER · Named entity recognition · Dataset · Labeled data · Cachaça

Token	Label
Descrição	O
da	O
Cachaça	O
:	O
A	O
Cachaça	O
Serra	B-NOME_BEBIDA
Limpa	I-NOME_BEBIDA
355 ml	B-VOLUME
é	O
uma	O
bebida	O
armazenada	O
por	O
seis	B-TEMPO_ARMAZENAMENTO
meses	I-TEMPO_ARMAZENAMENTO
em	O
tonéis	B-RECIPIENTE_ARMAZENAMENTO
de	I-RECIPIENTE_ARMAZENAMENTO
inox	I-RECIPIENTE_ARMAZENAMENTO
.	O

Reconhecimento de Entidades Nomeadas

Language Resources and Evaluation
<https://doi.org/10.1007/s10579-023-09665-0>

ORIGINAL PAPER



**CachacaNER: a dataset for named entity recognition
in texts about the cachaça beverage**

Priscilla Silva¹ · Arthur Franco¹ · Thiago Santos¹ · Mozar Brito² ·
Denilson Pereira¹

“De cor amarelo-palha, possui uma mescla de aromas de frutas cítricas, mel e baunilha. No paladar, traz um gosto doce, mas que aguça as papilas salgadas. Além disso, causa uma sensação picante e um pouco alcoólica na boca. Retro-gosto agradável e moderado.”

NER dataset in the beverage domain, and can be useful for other types of beverages with similar entity categories, such as wine and beer. We describe the process of data collection, creation of the dataset and its experimental evaluation. As a result, we created a dataset containing over 180,000 tokens labeled in 17 entity categories. The labeling obtained an agreement coefficient of 0.857 among the labelers, according to the Fleiss' Kappa metric, which is considered almost perfect. In our experimental evaluation, we obtained a micro-F1 value equal to 0.933 in the test set. The size of the dataset, as well as the result of its experimental evaluation, are comparable to other datasets in the Portuguese language, even though ours has a greater number of entity categories.

Keywords NER · Named entity recognition · Dataset · Labeled data · Cachaça

Chatbots



Chatbot para Pessoas Surdas



Tabela 1: Exemplo de Frases em Glosa - Dicionário INES

Português	Glosa
Agora todas as pessoas têm um computador.	AGORA TOD@ PESSOA TER COMPUTADOR.
Agora eu vou ao banco porque preciso pagar a conta de luz.	AGORA EU IR BANCO PRECISAR PAGAR LUZ.
Eu tenho um débito com o banco, para comprar um carro.	EU DÉBITO BANCO COMPRAR CARRO.
O dinheiro é importante porque precisamos comprar coisas.	DINHEIRO IMPORTANTE PRECISAR COMPRAR COISA.
Hoje à tarde farei um saque no banco.	HOJE TARDE EU IR B-A-N-C-O DINHEIRO SAQUE.
Hoje irei ao banco para verificar o saldo.	EU IR HOJE BANCO VERIFICAR S-A-L-D-O.

Consulta a BD usando a Ling. Natural



Table **singer** columns: Singer_ID | Name | Birth_Year | Net_Worth_Millions | Citizenship

Table **song** columns: Song_ID | Title | Singer_ID | Sales | Highest_Position

Result:

	Title	Name
1	Do They Know It's Christmas	Liliane Bettencourt
2	F**k It (I Don't Want You Back)	Liliane Bettencourt
3	Cha Cha Slide	Christy Walton
4	Call on Me	Iris Fontbona
5	Yeah	Christy Walton
6	All This Time	Gina Rinehart
7	Left Outside Alone	Jacqueline Mars
8	Mysterious Girl	Susanne Klatten

Consulta a BD usando a Ling. Natural

Performance evaluation of LLMs in the Text-to-SQL task in Portuguese

Blind Review

Resumo

Context: The rising need for consulting data in industry and academic contexts has fueled Text-to-SQL development, where natural language queries are translated into SQL, making data access easier. **Problem:** Most research focuses on English Text-to-SQL, leaving Portuguese—a language spoken by over 260 million people—underrepresented, creating challenges for organizations reliant on accurate data retrieval in Portuguese. **Solution:** This study evaluates the effectiveness of various Large Language Models (LLMs) on Portuguese Text-to-SQL tasks using a validated translation of the Spider benchmark. **IS Theory:** This research applies Task-Technology Fit (TTF) Theory to assess how well LLMs meet the needs of Portuguese Text-to-SQL tasks. TTF evaluates the match between LLM capabilities and task requirements, including language understanding, schema recognition, and SQL generation. Proper alignment is key for effective data retrieval, particularly for Portuguese-language applications in organizational decision-making. **Method:** A comparative analysis of seven LLMs—tested on both Portuguese and English Spider benchmarks—was performed. Exact Match (EM) and Execution Accuracy (EX) metrics measured performance, and a zero-shot prompting approach maintained consistency. **Results Summary:** Larger LLMs and specialized code models excelled, showing less performance variance between Portuguese and English tasks. Generalist models, however, produced verbose outputs, which may limit practical use in production systems. **Contributions and Impact on IS:** This research establishes baseline Portuguese Text-to-SQL metrics and insights into language adaptability in LLMs, offering guidance for organizations seeking Portuguese-language data solutions. By bridging language gaps, it advances data-driven practices and fosters growth in Portuguese-speaking regions.

ACM Reference Format:

Blind Review. 2018. Performance evaluation of LLMs in the Text-to-SQL task in Portuguese. In *Proceedings of Make sure to enter the correct conference title from your rights confirmation email (Conference acronym 'XX)*. ACM, New York, NY, USA, 9 pages. <https://doi.org/XXXXXXX.XXXXXXX>

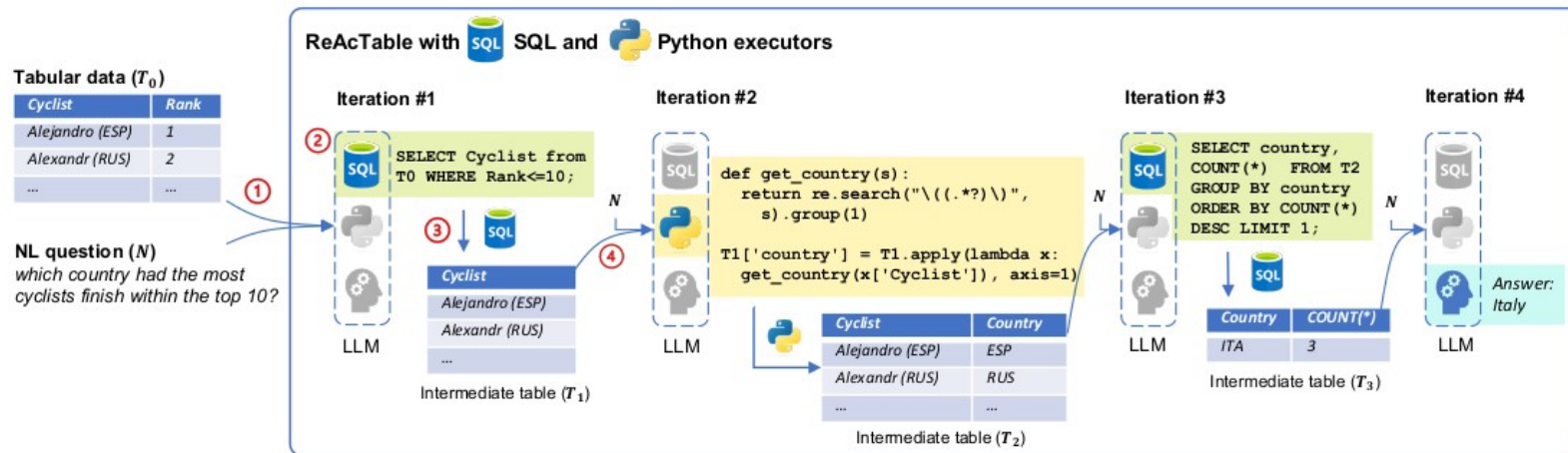
1 INTRODUÇÃO

A crescente demanda por acesso a dados na indústria e na academia tem impulsionado o desenvolvimento de novos meios para interação com bancos de dados. Neste contexto, a tarefa de *Text-to-SQL* emerge como uma solução promissora, permitindo que usuários formulem consultas em linguagem natural que são automaticamente convertidas em comandos na linguagem SQL [6, 20, 29]. Essa abordagem democratiza o acesso a dados estruturados, possibilitando que profissionais sem conhecimento técnico em SQL possam realizar consultas complexas de maneira eficiente.

Com o advento dos *Large Language Models* (LLMs), a tarefa de *Text-to-SQL* experimentou avanços significativos, alcançando níveis de acurácia anteriormente considerados inatingíveis. Modelos como GPT-4 demonstraram capacidade notável [6] em compreender nuances linguísticas e gerar consultas SQL precisas a partir de instruções em linguagem natural. No entanto, assim como em outras aplicações de processamento de linguagem natural, a maioria das pesquisas e desenvolvimentos nesta área tem se concentrado primariamente no idioma inglês [3], criando uma lacuna significativa para usuários de outras línguas.

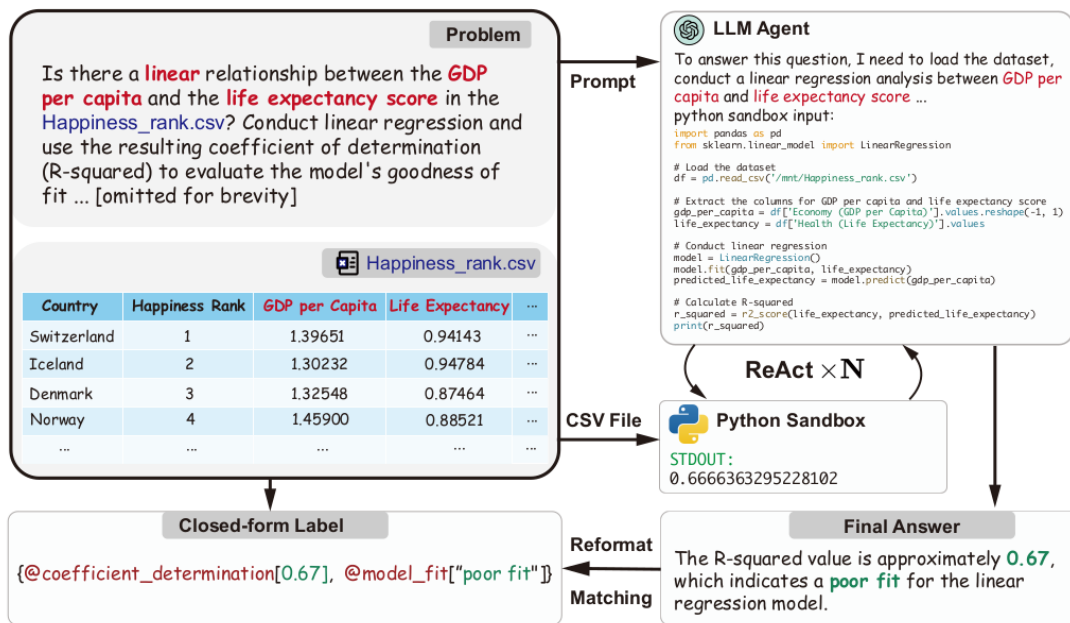
Para a comunidade lusófona, que engloba mais de 260 milhões de falantes em nove países, a disponibilidade de soluções *Text-to-SQL* em português representa não apenas uma questão de acessibilidade, mas também um imperativo para a democratização do acesso a dados. A necessidade de interagir com bancos de dados em língua

Consulta a BD usando Agentes



Referência: Zhang et al. ReAcTable: Enhancing ReAct for Table Question Answering. VLDB Endowment, 2024

Análise de Dados



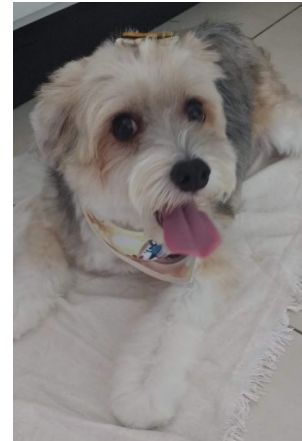
Referência: Hu et al. InfiAgent-DABench: Evaluating Agents on Data Analysis Tasks. arXiv, 2024



Aplicações em Processamento de Imagens

Aplicações em Processamento de Imagem

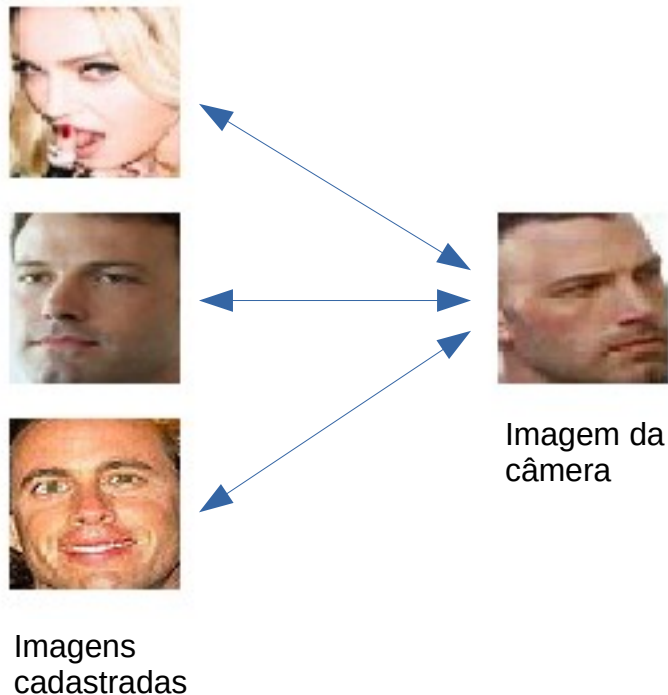
- ◆ Classificação de imagem
- ◆ Detecção de objetos
- ◆ Reconhecimento de faces
- ◆ Veículos autônomos



Cachorro? (S/N)



Reconhecimento Facial



Abordagem de função de similaridade

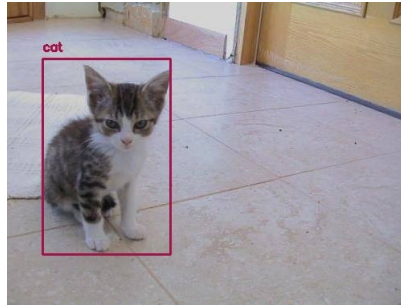
$d(\text{img1}, \text{img2})$ = o grau de diferença entre as imagens

If $d(\text{img1}, \text{img2}) \leq \alpha \implies$ mesma
 $> \alpha \implies$ diferente

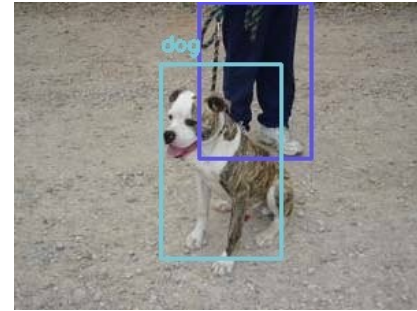
Detecção de Objetos



Classificação: “Gato”



Classificação com
localização: “Gato”



Detecção de Objetos:
Localiza e classifica
múltiplos objetos



Aplicações Híbridas

Extração de Informação de Formulários

Language Resources and Evaluation
<https://doi.org/10.1007/s10579-024-09802-3>

ORIGINAL PAPER



UFLA-FORMS: an academic forms dataset for information extraction in the Portuguese language

Victor Gonçalves Lima¹ · Denilson Alves Pereira¹

Accepted: 16 December 2024

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Abstract

Information Extraction aims to analyze and extract relevant information in document samples. For visual documents, such as academic and commercial forms, key-value pair extraction is capable of extracting and grouping the requested information automatically. The state of the art presents multimodal models capable of analyzing document's image, text and layout. To study and use these models in downstream tasks, labeled datasets must be available for fine-tuning of models. However, publicly available data is scarce, even more so for the Portuguese language. In this context, this work aims to provide a dataset of academic forms, named UFLA-FORMS, for the task of Information Extraction. The dataset is composed of 200 manually labeled samples, containing 7710 entities with 4442 relationships pairs between them. The labeling obtained a Kappa coefficient of agreement of 0.918 for the labeling of entities and 0.909 for the relationships attributed between them. The dataset was experimentally evaluated through cross-validation with hyper-parameter search in Named Entity Recognition and Relation Extraction tasks, obtaining, respectively, an average F_1 of 0.921 and 0.846.

Keywords Document intelligence · Information extraction · Dataset · NLP

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ANEXO 1

**FORMULÁRIO DE INSCRIÇÃO PARA CANDIDATURAS A
COMISSÃO DE COORDENAÇÃO GERAL DE CURSO DE PÓS-GRADUAÇÃO**

Nome dos professores que comporão a comissão geral de coordenação de curso:

Coordenador de curso: Henrique Mariano Cortés

Telefone: (35) 9734-2445

Localização na UFLA: Departamento de Estatística

SIAP: 567767673

Idade: 42 anos

Tempo de experiência na Educação a Distância: 2 anos

Tempo de Experiência em Magistério no Ensino Superior: 14 anos

Membro 1: Natália Clara de Ferreira

Telefone: (68) 93248-4223

Localização na UFLA: Departamento de Computação Aplicada

SIAP: 845998356

Membro 2: Sabrina Olívia Cortés Faria Nunes

Telefone: (83) 92526-3638

Localização na UFLA: Departamento de Educação em Ciências Físicas e Matemática

SIAP: 847504003

Fig. 7 Clipping of sample “072” from UFLA-FORMS. Highlighted yellow, blue, green and red entities represents, respectively, the Header, Question, Answer and Other classes. It is possible to observe the tree-shaped hierarchy formed by the relationships between the form’s entities.

Extração de Informação de Tabelas

Instância	S							
	Com busca local				Sem busca local			
	A	R	B	K	A	R	B	K
1	-5,25	3	-2,25	3	-2,25	-2,25	3	3
2	-35	25	5	-15	-25	15	35	-5
3	-66,5	57	-38	19	-9,5	-38	57	19
...	...	...	...	...	...	...	...	...

Figure 1: Example table

	Col. 1	Col. 2	Col. 3	Col. 4	Col. 5	Col. 6	Col. 7	Col. 8	Col. 9
Row 1						S			
Row 2	Instância		Com busca local				Sem busca local		
Row 3		A	R	B	K	A	R	B	K
Row 4	1	-5,25	3	-2,25	3	-2,25	-2,25	3	3
Row 5	2	-35	25	5	-15	-25	15	35	-5
Row 6	3	-66,5	57	-38	19	-9,5	-38	57	19
	...	...	...	...	...	...	...	...	...

Figure 2: Rows and columns of the example table

Extração de Informação de Tabelas

Instância	S							
	Com busca local				Sem busca local			
	A	R	B	K	A	R	B	K
1	-5,25	3	-2,25	3	-2,25	-2,25	3	3
2	-35	25	5	-15	-25	15	35	-5
3	-66,5	57	-38	19	-9,5	-38	57	19
...	...	...	...	...	...	...	...	...

Figure 3: Spanning cells in a table

Instância	S							
	Com busca local				Sem busca local			
	A	R	B	K	A	R	B	K
1	-5,25	3	-2,25	3	-2,25	-2,25	3	3
2	-35	25	5	-15	-25	15	35	-5
3	-66,5	57	-38	19	-9,5	-38	57	19
...	...	...	...	...	...	...	...	...

Figure 4: Header example

Extração de Informação de Tabelas

A Three-Dimensional Topological Positioning Proposal for the Table Recognition Task with Generative Models

Removed for double-blind review

Abstract

Context: The growing volume of data generated by information systems necessitates efficient indexing and searching of heterogeneous data. Tables in visual documents represent a significant form of semi-structured data that is valuable yet challenging for automated interpretation. Table Recognition is the task that aims to adapt tabular data for improved processing by automated systems.

Problem: Recent advancements in language processing, particularly through Large Language Models (LLMs), have influenced Table Recognition. However, many generative models still lack optimal adaptation for this task. Notably absent in the literature is the integration of topological data during the generation process.

Solution: To address this gap, we developed and evaluated a novel generative algorithm for Table Recognition that incorporates topological data.

IS Theory: This research is grounded in Information Processing Theory, focusing on enhancing data automation and processing.

Method: We adapted a pre-trained model to integrate topological data, such as table cell coordinates, during the generation process. We then compared this model to a baseline that did not use topological information.

Results Summary: The adapted model demonstrated improved performance in the task, achieving an 8.8% increase in Tree Edit Distance Score. However, it exhibited a higher tendency for repeating patterns and hallucinations.

Contributions and Impact to the Field of IS: This work paves the way for more task-specific generative models, potentially enhancing efficiency and accuracy of the Information Systems they support.

Keywords

Table Recognition, Visual Document Understanding, Generative Model, Information Processing

However, a significant challenge arises when it comes to the automated processing of tables found in visual documents. Tables are predominantly designed for human interpretation, requiring readers to possess a certain interpretive ability. This ability encompasses the recognition of symbols, language conventions, formatting, document style, and relationships between data points. For traditional computerized systems, this task proves non-trivial. Due to the vast diversity of styles, existing topologies, and potential data noise, the ability to comprehend tabular elements in visual documents remains elusive using solely formal rules.

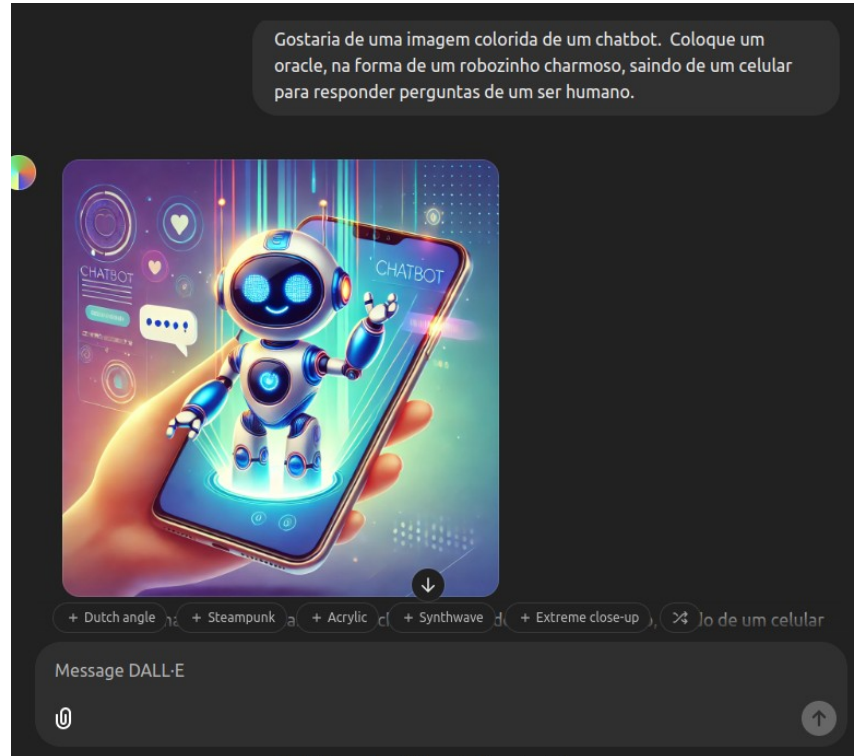
In the broader context of information processing within Information Systems, table processing serves as a pivotal element that connects raw data with meaningful interpretations. Within this framework, tables facilitate the organization and presentation of information, making complex datasets more accessible for analysis and decision-making. It enhances the capabilities of Information Systems by allowing for the extraction, transformation, and loading of tabular data into various analytical environments. This process is vital for converting visually represented information into formats that can be efficiently utilized for reporting, trend analysis, and strategic planning. As such, effective table processing not only improves data usability but also significantly amplifies the overall efficacy of Information Systems in delivering actionable insights to users. [14]

1.1 The Table Recognition Task

While there is not a universally accepted definition of how a table structure is defined, common patterns emerge. Various office applications (Microsoft Word, Google Docs, LibreOffice Writer), markup languages (HTML, LaTeX), and databases [2, 4, 18, 27, 28] share many commonalities on how to represent tabular data. These patterns provide valuable insights into the construction of tables in visual documents.

Utilizing the table in Figure 1 as an example, a visualization of

Geração de Imagem



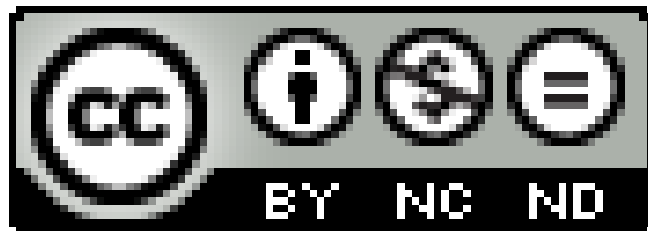


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